

Coherent-Offset Aliasing in UAV-to-Satellite Geo-Localization: Taxonomy, Measurement, and Why Consistency-Based Rejection Fails

[Author list pending]

Abstract—UAV geo-localization against satellite reference imagery fails on repetitive terrain in ways standard robustness tools cannot see. We characterize the failure on UAV-VisLoc, a published benchmark, and separate it into two classes: *whole-tile* aliases, where the matcher locks onto a look-alike tile hundreds of metres away, and *sub-tile* aliases, where the match lands on the correct tile but displaced by one period of a repeating structure invisible at satellite resolution. Three measurements make the case. First, signed per-frame offsets ($n = 610$ contiguous frames) show that magnitude histograms—the standard reporting convention—hide the alias class’s bimodality and its hole at zero. Second, a benchmark of four published rejection designs over one accepted-fix stream shows that sequential consistency and ORB-SLAM’s three-keyframe rule discriminate *backwards* on the sub-tile class (retaining wrong fixes at a higher rate than correct ones), while pairwise consistency and robust frame alignment attain forward discrimination only by keeping 7–40% of correct fixes; no consistency-based method meets a pre-registered adoption bar, whereas a prior-ratio gate separates the whole-tile class completely (0 of 7 fatal fixes kept per drift at 150 m and 300 m drift). Third, the alias offset is temporally coherent ($2.55\times$ below a shuffled null at lag 1)—which is exactly why consistency fails—and the mechanism-motivated countermeasure, a multi-hypothesis mixture filter, recovers none of the oracle gap at any prior quality because the alias lattice is per-field and unobservable in the satellite texture. Held-out regions and an independently collected urban dataset confirm the class is specific to repetitive farmland; an attempted replication on an agricultural-robot dataset is structurally impossible (no aerial imagery exists). We contribute (i) a taxonomy with per-class rejection behaviour, (ii) the first backwards-rate tables for published rejectors, (iii) a sign-folding diagnostic for error reporting, (iv) a falsified countermeasure with quantified oracle bounds (median $31.1\rightarrow 14.0$ m under the strongest available oracle), and (v) a pre-registered replication protocol.

Index Terms—Geo-localization, perceptual aliasing, robust estimation, UAV, satellite imagery, failure analysis.

I. INTRODUCTION

GPS-denied UAV navigation increasingly relies on matching nadir drone imagery against georeferenced satellite tiles to anchor drift-prone visual-inertial odometry. Published systems report median errors of metres on favourable terrain [1]–[3]. Failure analysis, however, is thin: benchmarks report accuracy at thresholds ($A@X$ m) over pooled terrain, which is blind to how, where, and why the tail fails.

We document a failure class on repetitive terrain—farmland with parallel furrows, forest canopy—where accepted matches are wrong by tens to hundreds of metres *with full geometric and appearance support*. We show this class defeats the standard robustness toolbox for a measurable, mechanical reason, and we quantify what would be required to fix it.

Coherent (mutually consistent) outliers in SLAM are studied theoretically for indoor scenes [4]; row-level aliasing on vineyard rows is reported from ground robots with LiDAR [5]–[7]. Neither provides cross-view (nadir UAV $\leftrightarrow$ satellite) measurements, per-class rejection behaviour, nor the temporal-coherence and appearance-optimality measurements we report. STHN notes self-similar patterns as a confound in UAV thermal $\leftrightarrow$ satellite homography [8] but does not analyze them. Robust pose-graph back-ends (PCM, GNC, TACO) are designed against exactly this outlier shape [9]–[12]; we measure how the standard representatives of this family behave on real cross-view data.

Contributions.

- 1) **A taxonomy** separating whole-tile from sub-tile aliases, each with a distinct geometry, temporal signature, and rejection behaviour (Sec. IV-A, IV-C).
- 2) **Backwards-rate tables** for four published rejection methods over a shared fix stream: sequential consistency and the three-keyframe rule keep wrong fixes at a *higher* rate than correct ones; PCM and robust frame alignment discriminate forward only by destroying 60–93% of correct fixes; only a prior-normalized distance gate works, and only on the whole-tile class (Sec. IV-B).
- 3) **The sign-folding diagnostic**: absolute-value error histograms fold symmetric alias structure into an apparently continuous distribution; signed offsets recover the axis and the hole at zero (Sec. IV-A).
- 4) **A falsified countermeasure**: the mixture hedge filter that follows from the mechanism recovers none of the oracle gap, because the alias lattice is per-field and unobservable—with a reproducible oracle bound of 14.0 m median quantifying the residual (Sec. IV-D).
- 5) **Held-out and cross-dataset controls** confirming the class is terrain-specific, plus a pre-registered replication protocol (Sec. IV-E, IV-F).

II. RELATED WORK

UAV–satellite geo-localization. Retrieval-based systems [1], [13], [14] and geometric pipelines (homography, as in the production matcher studied here; PnP against DSM [2], [3]) are benchmarked by pooled accuracy-at-threshold. AnyVisLoc’s Table 6 is representative: its baseline scores 74.1% A@5 m against an aerial photogrammetry reference but 18.5% A@5 m against a satellite reference [1]—the satellite-reference tail exists, is large, and is not analyzed per-region. Our work is complementary: we analyze *why* that tail exists rather than improving its head.

Perceptual aliasing and robust estimation. Aliasing in place recognition is a classic concern [15]; sequence-based VPR exploits temporal consistency to combat it [16]. Modern robust back-ends address mutually consistent outliers with pairwise consistency (PCM [9], Kimera-RPGO [10]), graduated non-convexity (GNC [11]), or robust kernels on pose-graph factors (TACO [12]). Lajoie et al. analyze coherent outliers in indoor SLAM theoretically and note that mutually consistent outliers defeat initial-guess-dependent rejection [4]. Vineyard robotics reports row-level aliasing from ground LiDAR qualitatively [5], [6]. Multi-hypothesis handling of aliasing dates to hypothesize-and-verify loop closure [17]. None of these measures the rejection *rate* of published methods on a shared cross-view fix stream, nor the temporal coherence of the alias offset itself—the measurements we contribute.

Error distribution analysis in VPR. Reporting conventions use magnitude histograms and CDFs. We show a class of symmetric bimodal error—the signature of period-quantized aliasing—is invisible in magnitude form, and propose signed-offset reporting as a diagnostic (Sec. IV-A).

III. METHOD

A. System under study

The pipeline is a production map matcher: ORB+AKAZE+SIFT pooled correspondences, MAGSAC homography [18], image-centre projection through the inverse homography, patch-wide NCC verification at 0.30, `min_inliers`=10, DEM-corrected AGL scale. Candidate generation is prior-centred (5×5 tile ring). Frames carry a drifted prior injected as a random walk at 150/300/600 m. Dataset: UAV-VisLoc [13], eleven Chinese sites; all measurements use the dataset’s drone GPS as truth.

B. Measurement protocol and definitions

Error classes. A fix is *good* if its error against the dataset GPS is < 20 m, *mid* if $20\text{--}50$ m, and *fatal* if ≥ 50 m. These boundaries are fixed a priori and used across all experiments.

Signed offsets. For each frame, we match against the ground-truth tile and record the signed (north, east) offset between the recovered position and the dataset’s GPS record. Frames are *contiguous* in flight order ($n = 610$ solved on R04), because step-sampling destroys the temporal structure the coherence question needs.

Fix stream. For rejection benchmarks, we run the production matcher under injected drift and keep every accepted fix

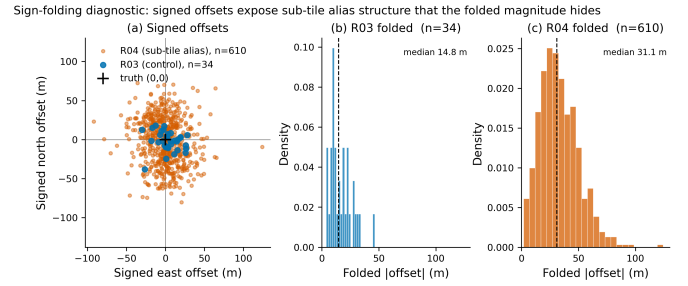


Fig. 1. The sign-folding diagnostic. Left: signed (north, east) offsets on R04 ($n = 610$ contiguous) and R03 (step-sampled). The R04 alias class is axis-aligned with a hole at zero; R03 is unstructured. Right: the same R04 data folded to magnitudes—the standard reporting convention—appears continuous and hides the alias class.

with its estimate, prior, and prior uncertainty ($n = 40$ per region per drift, seed fixed).

Metrics. Good-kept% / fatal-kept% with denominators in-line; discrimination ratio = good-kept/fatal-kept (> 1 = forward discrimination). Adoption bar for a rejector: fatal cut $\geq 25\%$ while keeping $\geq 80\%$ of good fixes (pre-registered before any benchmark ran).

C. The rejector benchmark

Four published designs are implemented over the shared fix stream, using only deployed signals (priors, estimates, filter-reported uncertainty): sequential consistency (a fix survives if a neighbouring fix agrees within a motion-compensated tolerance; the ORB-SLAM3 covisibility principle [19]), ORB-SLAM’s three-consecutive-keyframe rule [20], PCM-style greedy maximum clique [9], and VINS-Mono-style robust frame alignment (median prior $\leftrightarrow$ estimate offset, reject residual $> \tau$) [21]. The prior-ratio gate (fix kept iff distance to prior $\leq 1.5 \times$ the prior’s uncertainty) is evaluated in deployed (filter RMS) and oracle (true prior error) forms.

IV. EXPERIMENTS

A. Error structure: the taxonomy and the sign-folding diagnostic

Setup. Signed offsets on GT tiles, R04 ($n = 32$ step-sampled for regional statistics; $n = 610$ contiguous for the full distribution) and R03 ($n = 30/557$) as control. The $n = 32$ sample and the $n = 610$ stream are *different* samples of the same region; baseline medians therefore differ (32.7 m on $n = 32$, 31.1 m on $n = 610$) and are always quoted with their sample size.

Result 1—magnitude histograms hide the alias class. The signed R04 offsets are axis-aligned and bimodal: axial resultant $R = 0.50$ (vs. 0.19 control) with the *hole at zero* (6/32 frames within ± 10 m along the dominant axis, vs. 14/30 on the control). Folded to magnitudes, this is exactly the “continuous, no-gap” distribution that earlier analysis misread as imprecision (Fig. 1).

Result 2—the axis is per-field, not per-region. At $n = 32$ the offsets split into two orientation groups (bearings $145\text{--}193^\circ$ and $279\text{--}357^\circ$): the flight crosses fields whose furrow

TABLE I

REJECTION RATES OVER THE SHARED ACCEPTED-FIX STREAM
(GOOD-KEPT%/FATAL-KEPT%; DISCRIMINATION RATIO >1 IS FORWARD).
TOL= $\tau=100$ M THROUGHOUT; PRIOR-RATIO THRESHOLD $r=1.5$; BOLD
CELLS PASS THE PRE-REGISTERED ADOPTION BAR (FATAL CUT $\geq 25\%$,
GOOD KEPT $\geq 80\%$).

Rejector	d150	d300	d600
Seq. consistency	0.96 (96/100)	0.84 (84/100)	0.81 (59/73)
3-consecutive	0.95	0.97	0.45 ($n=3$)
PCM max-clique	3.02 (40/13)	3.17 (23/7)	0.79
Frame alignment	2.09 (56/27)	2.80 (20/7)	1.12
Prior-ratio (RMS)	2.12 (99/47)	1.08	1.00
Prior-ratio (oracle)	2.31 (92/40)	1.95 (97/50)	1.57 (100/64)

orientation changes. A single-region-axis oracle degrades accordingly: the sub-tile snap oracle (best integer period removed along one axis) falls from $2.39\times$ over a random-axis null at $n = 16$ to $1.72\times$ at $n = 32$, while its absolute ceiling stays stable (12.0–12.7 m median against a 32.7 m baseline). The alias is a lattice structure *per field*.

Result 3—the wrong lock is the appearance optimum, unanimously. At $n = 32$, masked patch NCC over displaced homographies selects displacement $k = 0$ on 32/32 frames; the oracle displacement recovers 20.4 m of median error (32.7→12.3 m). Appearance is not merely uninformative on this class; it is monotone *against* the truth.

Result 4—whole-tile aliases are a different class. On forest (R06), the error distribution is bimodal with a clean 40–150 m gap: discrete jumps to look-alike tiles 1–2 tile widths along track. Their offsets are coherent across the flight turn, following the aircraft.

B. The backwards-rate benchmark

Setup. Production matcher, regions R03 (control, 0 fatal), R04 (sub-tile), R06 (whole-tile); drifts 150/300/600 m; 255 pooled accepted fixes. Four rejectors plus two prior-ratio forms.

Finding 1—sequential consistency and the three-keyframe rule discriminate backwards on the sub-tile class. They keep wrong fixes at a *higher* rate than correct ones at every drift (ratio ≤ 0.97 wherever n permits); at tol=100 m, 100% of fatal fixes survive at d150 and d300 (Fig. 2).

Finding 2—the forward ratios that do exist are coverage-destroying. PCM and frame alignment reach ratios of 2–3 at tol=100 m only by keeping 7–40% of correct fixes—the purity-for-coverage trade, measured as a rate. No consistency-based method passes the adoption bar in any cell at any drift.

Finding 3—the whole-tile class is fully separable, the sub-tile class is invisible, per region. On R06, the prior-ratio gate with honest uncertainty keeps 100% of good fixes and 0 of 7 fatal fixes per drift at d150 and d300 (14 fatal fixes pooled over the two drifts). On R04 it scores 0.93–1.00: a 20–80 m alias is small against a 300 m prior, so no ratio threshold can see it. On the R03 control, the consistency filters destroy 15–76% of good fixes—collateral measured, not asserted.

Rejection benchmark: good-kept vs fatal-kept rate per method, green = forward discrimination ratio (good/fatal > 1)

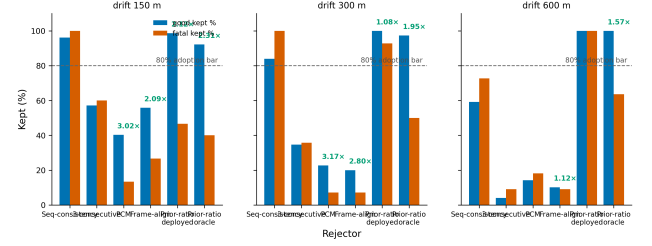


Fig. 2. Backwards-rate benchmark over the pooled fix stream (255 accepted fixes). Bars: good-kept vs. fatal-kept percentage per rejector and drift; dashed line: the 80% good-kept adoption bar. Sequential consistency and the three-keyframe rule keep fatals at $\geq$ good rates (ratio ≤ 1); PCM and frame alignment score >1 only by destroying coverage.

TABLE II

MEDIAN PAIRWISE OFFSET DIFFERENCE VS. FRAME LAG (M). NULL: OFFSETS SHUFFLED WITHIN GROUP (10 TRIALS). $n =$ ALIAS PAIRS.

lag	n	alias: med / null	good: med / null
1	25	27.5 / 70.1	2.55$\times$ below null
2	17	46.8 / 86.1	1.84 $\times$
3	15	27.9 / 63.9	2.29 $\times$
5	12	80.0 / 90.1	1.13 $\times$
8	11	60.3 / 93.5	1.55 $\times$

C. Temporal coherence: the mechanism

Setup. Truth-referenced signed offsets, contiguous frames, R04 $n = 610$ (alias ≥ 50 m: 93; mid 20–50 m: 374; good < 20 m: 143), R03 control $n = 557$. Per-lag pair counts inline.

Finding 4—the alias offset is locally coherent and decays with lag. Adjacent alias frames differ by 27.5 m where random pairing gives 70.1 m; the advantage decays to $\sim 1.1\times$ by lag 5–8 (Fig. 3). The lock is not rigid: it drifts slowly (the period index changes occasionally), coherent over 2–4 frames. Correct fixes are independent per frame (at null at every lag, 0.78–1.13 $\times$) and the R03 control matches them. The 20–50 m band is *partially* coherent—a mixture of alias and noise—which explains why threshold rejectors tuned anywhere in that band fail (Finding 1).

Consequence. Within the coherence window, an alias chain is exactly as motion-consistent as the truth: pairwise consistency has a gauge symmetry (constant offsets cancel in frame differences), so PCM/sequential consistency are provably blind, not merely weak. Beyond the window, the offset has drifted enough to survive any tolerance that keeps good fixes (their own ~ 17 m scatter). The backwards rates of Sec. IV-B are the statistical image of this geometry.

D. The countermeasure fails for a measured reason

Setup. Two routes: (a) estimate the alias lattice from the map itself—detrended autocorrelation of a 500-px crop around each locked position, strongest local maximum in the 9.5–95 m band, $n = 610$; (b) a multi-hypothesis mixture filter: candidate true positions on a $k \in [-4, 4]$ lattice (period 20 m, axis 171° , from Sec. IV-A), posterior from the prior likelihood, output = posterior mean; prior uncertainty swept 5–300 m in random-walk and IID forms. Kill criterion pre-registered: posterior-

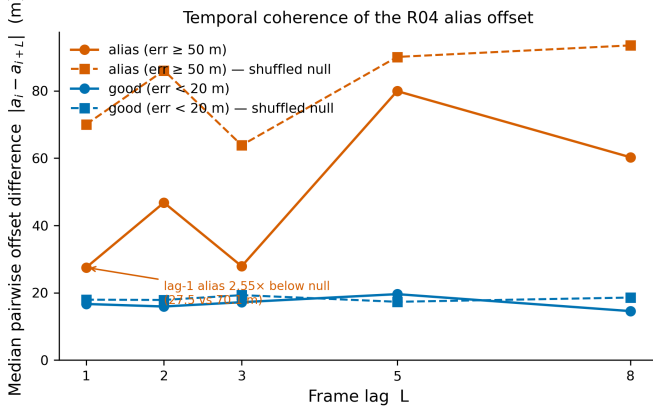


Fig. 3. Temporal coherence of the alias offset (R04, $n = 610$ contiguous). Solid: median pairwise offset difference within the alias (≥ 50 m) and good (< 20 m) groups; dashed: shuffled nulls. The alias group sits $2.55\times$ below its null at lag 1 and decays to $\sim 1.1\times$ by lag 5–8; correct fixes are independent at every lag.

mean output must recover $\geq 30\%$ of the oracle gap ($17.1\text{ m} \Rightarrow \geq 5.1\text{ m}$) at prior $\sigma \leq 20\text{ m}$.

Result (a)—the lattice is not in the map. In-band periodicity is found on 9–17% of frames (lowest for the alias group) and never aligns with the alias offset direction (median angular error $44\text{--}52^\circ$, uniform). At 1.19 m/px the satellite texture does not expose the furrow lattice the matcher locks onto.

Result (b)—the mixture recovers nothing; MAP is actively harmful. Baseline $k = 0$ median 31.1 m ($n = 610$); single-axis oracle 14.0 m (gap 17.1 m). Posterior-mean output moves $-6.1 \dots +0.1\text{ m}$ across all prior forms and qualities (worst at $\sigma = 50$ IID: -6.1 m , far below the $\geq 5.1\text{ m}$ bar); MAP output degrades to 68.4 m at $\sigma = 300\text{ m}$. The failure is mechanical: frame differences cannot see a constant offset (Sec. IV-C), the prior is the only anchor, and the true offsets are not multiples of any single global lattice—the per-field revision of Sec. IV-A at stream scale.

Finding 5—recovery requires the per-field lattice, which no deployed signal provides. The failure chain is closed: the alias offset is quantized along a field axis (Sec. IV-A), the axis is not in the satellite texture (a), frame differences cannot see a constant offset (Sec. IV-C), and no prior tightness restores the modes (b). The 14.0 m oracle is the non-degenerate bound for any matcher-side fix; the IID $\sigma = 5$ mixture—the video-rate odometry limit—extracts nothing from it (31.2 m).

E. Held-out controls and the terrain boundary of the class

The sub-tile coherent-alias class appears only on repetitive furrow farmland (R04). Desert and clean farmland are control terrain; the mountain plateau and the two additional held-out regions (R02, R10) belong to the *unsolvable class* (a 0–5% inlier ceiling even against the region’s own ground-truth tile), which is a different, no-fix failure mode rather than a wrong-fix one.

F. External replication: attempted and executed

Attempted replication 1—TEMPO-VINE (rejected on structure). The most promising published agricultural dataset

TABLE III
TERRAIN BOUNDARY OF THE ALIAS CLASS. PRODUCTION MATCH:
DRIFTED-PRIOR SOLVE RATE, $n = 40$ AT D300. GT-TILE SOLVE:
CONTIGUOUS FRAMES AGAINST THE REGION’S OWN SATELLITE TILE.

Region	Terrain	Prod. match (d300)	GT-tile solve
R03	farmland	34/40, 0 fatal	557, good at null
R04	furrow farmland	alias source	610, alias class
R05	mountain plateau	1/40	5/100 (5% ceiling)
R11	desert	34/40, 1 fatal	199: 8 alias (4%)
R02	farmland/river	—	1/40 usable
R10	orchard hills	—	0/40 usable

TABLE IV
AERIALVL REPLICATION (PROTOCOL STEPS 1–3).

Metric	Seq 03-11	Seq 03-16
Solved frames (≥ 15 inliers)	94/200 (47%)	0/200 (0%)
Hole at zero	5.3%	—
Axial resultant $R(180^\circ)$	0.70	—
Anisotropy (along/perp)	$9.7\times$ (16.3/1.7 m)	—
Median error	17.1 m	—
Median offset vector	(-16.0 N , $+4.1\text{ E}$) 16.5 m	—
Good-group lag-1 coherence	$0.59\times$ of null	—
Error groups (good/mid/alias)	75 / 15 / 4	—

for this mechanism, TEMPO-VINE [7], provides RTK-grade GPS truth ($1\text{--}2\text{ cm}$) in trellis and pergola vineyards—but its imagery is a forward-oblique RealSense D435 RGB-D camera on a ground rover at $\sim 1\text{ m}$ height. Nadir aerial imagery matchable against satellite ortho does not exist in the dataset, so protocol requirement 1 fails and no measurement is possible. This is a *structural* absence in the field: no public dataset currently pairs repetitive-crop aerial imagery with satellite references and continuous GPS.

Executed replication 2—AerialVL (class absent; protocol guard fired). We ported the full protocol to AerialVL, an independently collected UAV dataset (HuggingFace hmf21/AerialVL, cc-by-4.0, no paper; per-frame GPS in the image filenames; Qingdao, China; urban/coastal terrain) [22]. Two flight sequences were ingested (1,423 frames), the reference map built from the dataset’s georeferenced ortho (0.959 m/px), and protocol Steps 1–3 run.

Two findings follow. First, the alias class is **absent**: the alias tail is 4 frames (4.3%), too thin for any coherence measurement. The sign-folding signatures that *would* replicate (hole 5.3%, R 0.70, anisotropy $9.7\times$) are produced by a different mechanism—a constant 16.5 m common-mode georeferencing offset between the dataset’s ortho and its drone GPS (spread 3.1 m ; a cross-validated bias correction removes 14 m : $17.3 \rightarrow 3.3\text{ m}$). A constant offset is trivially “perfectly coherent”, so this is a false-positive source for the diagnostic—and a live corroboration of Sec. IV-C’s gauge symmetry: a constant offset is invisible to any consistency-based rejection. Second, the protocol’s own control guard fired: the *good* group is temporally coherent ($0.59\times$ of its null; a clean dataset requires $0.8\text{--}1.25\times$), meaning this dataset’s GPS truth is temporally correlated—a benchmark-hygiene property worth reporting on its own, and a reminder that coherence checks must control for truth-correlation before interpreting alias

coherence.

Follow-up. The one dataset class that could replicate the mechanism—repetitive-crop aerial imagery (the recently released ViLD dataset of UAV-to-satellite image matching over French fields, WACV 2026, access-gated [23])—is stated as future work with the pre-registered protocol.

V. DISCUSSION

What the field should take. (i) Pooled A@X m metrics on repetitive terrain hide a failure class with full geometric support; signed-offset reporting is a cheap diagnostic that exposes it. (ii) Robust-estimation defaults (PCM/GNC/sequential consistency) do not transfer to cross-view geo-localization on repetitive terrain: they fail backwards or at coverage-destroying purity, at measured rates. (iii) The fix that works for one alias class (prior-ratio for whole-tile) does not exist for the other; hedging (posterior-mean) is safe but inert, mode-commitment (MAP) is dangerous (68.4 m at $\sigma = 300$). (iv) An independent replication attempt shows the diagnostic’s false-positive mode (constant georeferencing bias) and a truth-correlation guard that any future replication must pass.

Limitations. Dataset GPS truth (~ 10 m class) bounds all absolute numbers—the R03 floor of ~ 13 m is plausibly the dataset’s own noise floor, not the matcher. Sample sizes: per-region fatal counts are 5–14 per drift; all denominators are reported inline, and the key measurements ($n = 32$ unanimity, $n = 610$ coherence) do not rest on small cells. The sub-tile class is measured on a single region (R04) and does not replicate on clean or unsolvable terrain; an external dataset with repetitive-crop aerial imagery does not exist publicly. All rates are measured with one matcher (ORB+AKAZE+SIFT+MAGSAC); the coherence mechanism (Sec. IV-C) is matcher-independent, but rate magnitudes are matcher-conditional. AerialVL’s georeferencing bias and correlated truth were discovered, not controlled for, in that replication.

VI. CONCLUSION

We characterized a failure class in UAV-to-satellite geo-localization that consistency-based robustness tools cannot see: coherent-offset aliases, split into whole-tile (separable by a prior-ratio gate) and sub-tile (unreachable by any measured signal) classes. The sub-tile class defeats sequential consistency and the three-keyframe rule *backwards* at measured rates, defeats PCM and frame alignment through coverage destruction, hides in magnitude histograms, and resists even the mechanism-motivated multi-hypothesis countermeasure, with a non-degenerate oracle bound of 14.0 m median on the worst region against a 31.1 m production baseline ($n = 610$). The class is terrain-specific: absent on desert, clean farmland, mountain plateau, unsolvable regions, and an independent urban dataset. The paper contributes the taxonomy, the rate tables, the sign-folding diagnostic, and the quantified boundary of what map matching can and cannot recover—and, by exclusion, directs future work to per-field map annotation or odometry-quality regimes we quantify as unreachable at 7 s frame spacing.

DATA AND CODE AVAILABILITY

The measurement harness, rejector benchmark, coherence analysis, and replication scripts, together with all per-frame result artifacts backing every table and figure, are available from the authors; the replication protocol is fully specified (five steps, exact commands, pre-registered kill criteria) so that all experiments can be re-executed end to end. UAV-VisLoc is public [13]; the AerialVL sequences used are public under cc-by-4.0 [22].

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